# ECE1508 Group Project

Enoch Lo
Farnaz Moayyedian
Chen Jiarong Edwin

October 2026

## 1 Abstract

Conventional controllers for robotic platforms are typically based on analytical models that do not capture dynamics that are difficult to model accurately (residual dynamics) such as friction and actuator dynamics. This often results in controllers having to be manually tuned and deployed with conservative safety margins to compensate for the residual dynamics, sacrificing performance for safety. Learning-based approaches can replace manual tuning and recover some performance by compensating for residual dynamics, but end-to-end reinforcement learning (RL) is sample-inefficient and does not take advantage of the structure from incomplete but partially representative analytical models. This project aims to combine both approaches by retaining a nominal model-based controller and using RL to learn only the residual dynamics. Restricting learning to the residual reduces the search space and improves sample efficiency. The learned residual is then used to correct the nominal model's predictions, producing a hybrid model that can relax safety margins. We will evaluate this hybrid approach on a pair of drones flying in proximity against the nominal controller and a pure RL baseline. The evaluation will be based on a subset of tracking error, control effort, model prediction accuracy, constraint violations, and real-world training time.

## 2 Key Outputs

1. Policy optimisation pipeline for pure RL model
2. Policy optimisation pipeline for residual model
3. Hybrid control framework combining the nominal and learnt residual models
4. Evaluation pipeline in Isaac Sim/gym-pybullet which serves as a stand-in for the real system, with ablations of the residual
5. Comparison between nominal controller, pure RL, and hybrid models

# 3 Requirements and Plan

1. Software: PyTorch, Isaac Lab, ProxFly's public code [1]
2. Hardware: GPU machine for running training/simulations in Isaac environment
3. Data: all experience generated in simulation
4. Preparation plan:
    - Oct 15 - 22: Simulation and training environment setup and PID control baseline
    - Oct 22 - 29: Downwash and ground-effect modelling in simulations
    - Oct 29 - Nov 5: Replicating ProxFly results using domain randomisation
    - Nov 5 - 12: Residual PPO for disturbance estimation / residual control policy
    - Nov 12 - 19: Refinement of training and disturbance compensation techniques
    - Nov 19 - 26: Circular tracking and aerial effect testing simulations
    - Nov 26 - Dec 10: Comparative studies against pure RL and ablation studies
    - Dec 10 - 17: Report and presentation preparations

# References

[1] R. Zhang, D. Zhang, and M. W. Mueller, "Proxfly: Robust control for close proximity quadcopter flight via residual reinforcement learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 13 683–13 689.